

Brunswick Raw Dataset Analysis

File: Brunswick_orig_1500ms_V2.hdf5 | Size: 15.57 GB | Raw Structure & Health Check

4,496,540

TOTAL TRACES

751

SAMPLES/TRACE (1500 MS)

1,817

SHOT GATHER COUNT

108

HDF5 OBJECT COUNT

Data Hierarchy & Configuration

Main Group: TRACE_DATA/DEFAULT

The full internal structure of the HDF5 file was analyzed prior to feeding the dataset into the U-Net neural network. It contains over 100 header fields resulting from vendor SEGY export.

Field Name	Format	Shape / Range
data_array	float32	(4,496,540, 751) [13.5 GB]
OFFSET / FLT	int32 / float64	20m – 71,890m
SHOTID / RECORDNUM	uint32	1 – 1,817
CDP / CDPTRACE	uint32	10001073 – 12351395
SAMP_RATE	uint32	2000 μ s (2 ms)

Data Quality & First-Break Fields

- FIRST_BREAK_TIME:** Value range [0.0, 0.0] — First break arrival times are completely empty in this raw file.
- MODELLED_BREAK_TIME:** Value range [0.0, 0.0] — Model picks are not present.
- Trace Density:** Average of ~2,474 traces per shot gather (min 1, max 3,358 traces/shot).

Findings & Risks

Main Issue: The raw Brunswick_orig_1500ms_V2.hdf5 file is unusable directly for U-Net model training due to missing first-break labels and heterogeneous trace counts per shot.

Required Action Steps:

- Trace sorting based on coordinates and offset values.
- Integration of first-break labels/masks from external sources.
- Standardization and filtering of shot gather density.
- Amplitude normalization and gain adjustment.

Technical Specifications

Matrix Shape	2D Segmentation (Traces $\times$ Time)
Disk Size	15.57 GB (data_array: 13.51 GB)
Coordinate System	X: 2,820,510–2,876,720 Y: 52,524,480–52,594,980

Preprocessed Dataset Analysis Report

Data Pipeline Prepared for U-Net Segmentation Framework

Filtered

DATA HYGIENE

Standardized

TRACE DIMENSION & GRID

3-Class

FIRST-BREAK MASK

U-Net-Ready

MODEL READINESS

Preprocessing Strategy

The raw dataset was preprocessed into 2D matrices suitable for U-Net image segmentation through the following steps:

- Shot Gathering:** Traces grouped by SHOTID to form structured 2D image matrices.
- Offset Ordering:** Traces sorted sequentially according to offset distances within each shot.
- Amplitude Scaling:** RMS normalization and Min-Max clipping applied across trace channels.
- Label Construction:** First-break masks created corresponding to picked arrival times.

Model Configuration & Segmentation

Pipeline Compatibility: Preprocessed datasets align directly with the `first_breaks_picking` framework for 2D visual segmentation.

Next Steps (Model Training):

- Splitting dataset into Train / Validation / Test sets (80/10/10).
- Data augmentation (Random flips, Gaussian noise injection).
- Loss function configuration using Combined Cross-Entropy + Dice Loss.

Pipeline Data Feasibility

Stage	Raw (Brunswick)	Preprocessed
First-Break Picks	Empty (0.0 ms)	Populated / Extracted
Trace Dimensions	Variable (1-3,358)	Fixed / Padded
Signal-to-Noise	Raw / Unfiltered	Bandpass / Normalized
Mask Classes	None	Before / Break / After

Summary Finding

The preprocessed dataset resolves the key limitations of the raw Brunswick dataset (empty break times and irregular dimensions), rendering it directly transferable into neural network tensors.